Larger-cohort solver validation

Correctness, candidate-only recovery and matched baseline costs

AutoRho, Bernstein-Lange and BSGS each returned checked answers for 768/768 targets. The candidate-then-rho solver also recovered all 768; the frozen prefix alone supplied 3. This report separates working correctness and measured baseline cost from novelty.

Subgroup bits	Curves	Targets	Candidate hits	Hybrid solved
12	12	192	2	192
14	12	192	1	192
16	12	192	0	192
18	12	192	0	192

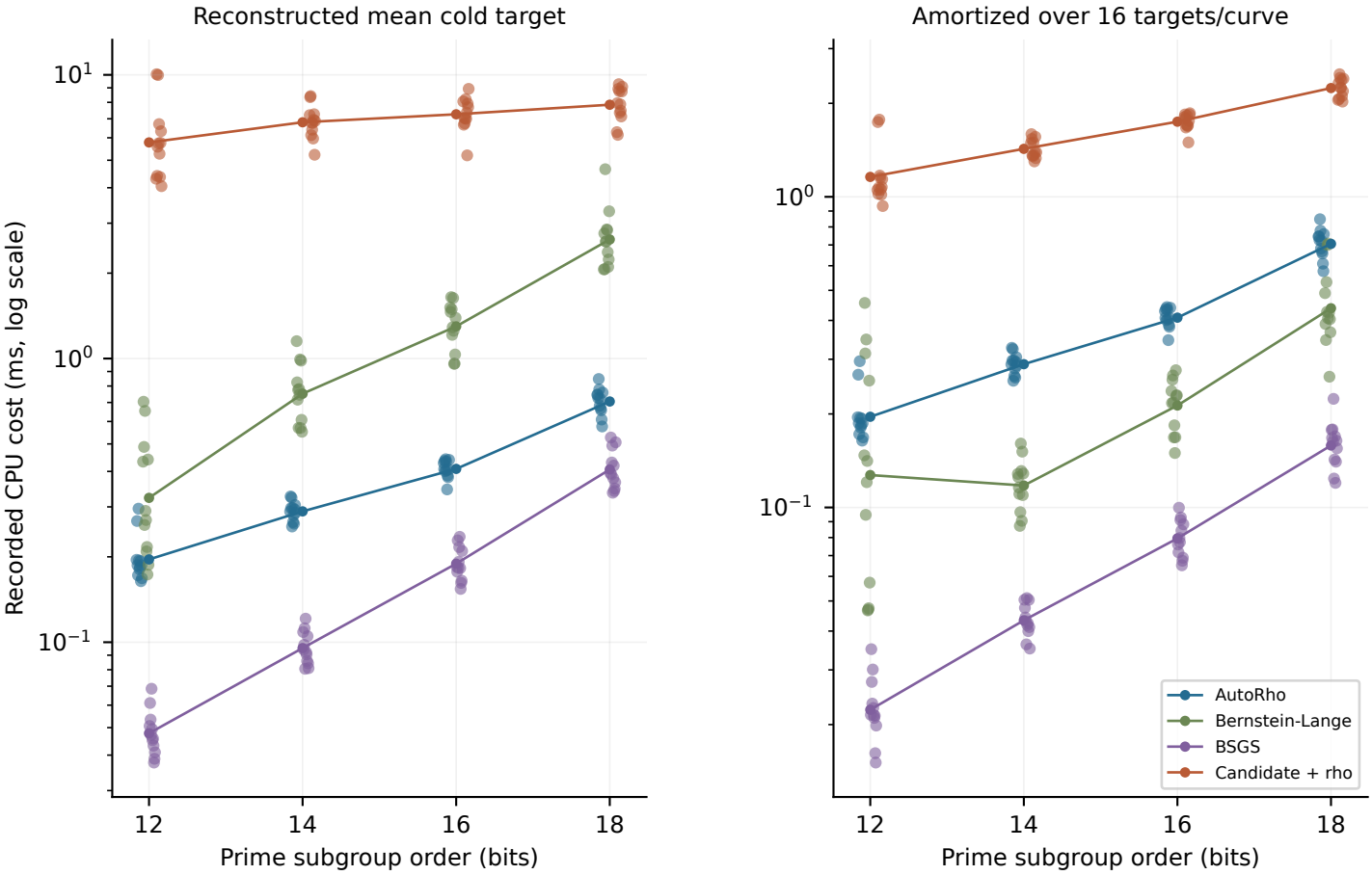
Arm	Solved	Failed	Censored	Invalid
AutoRho	768	0	0	0
Rho A/A	768	0	0	0
Candidate + rho	768	0	0	0
Bernstein-Lange	768	0	0	0
BSGS	768	0	0	0

Each arm requests 768 targets. The primary bounded candidate supplies 3 answers; the hybrid invokes rho 765 times. A complete hybrid answer via rho is not a candidate-only recovery. Rho A/A repeats the same solver and seed rather than adding a fifth independent algorithm.

The cohort has six curves per size/stratum cell. Full reuse is 16 targets per curve. The fixed 192-target core uses eight targets on 24 of these curves; its 576-target extension and full panel overlap. Curves are the replication units.

CPU cost at the tested sizes

Required setup and target work; operation counters retained separately



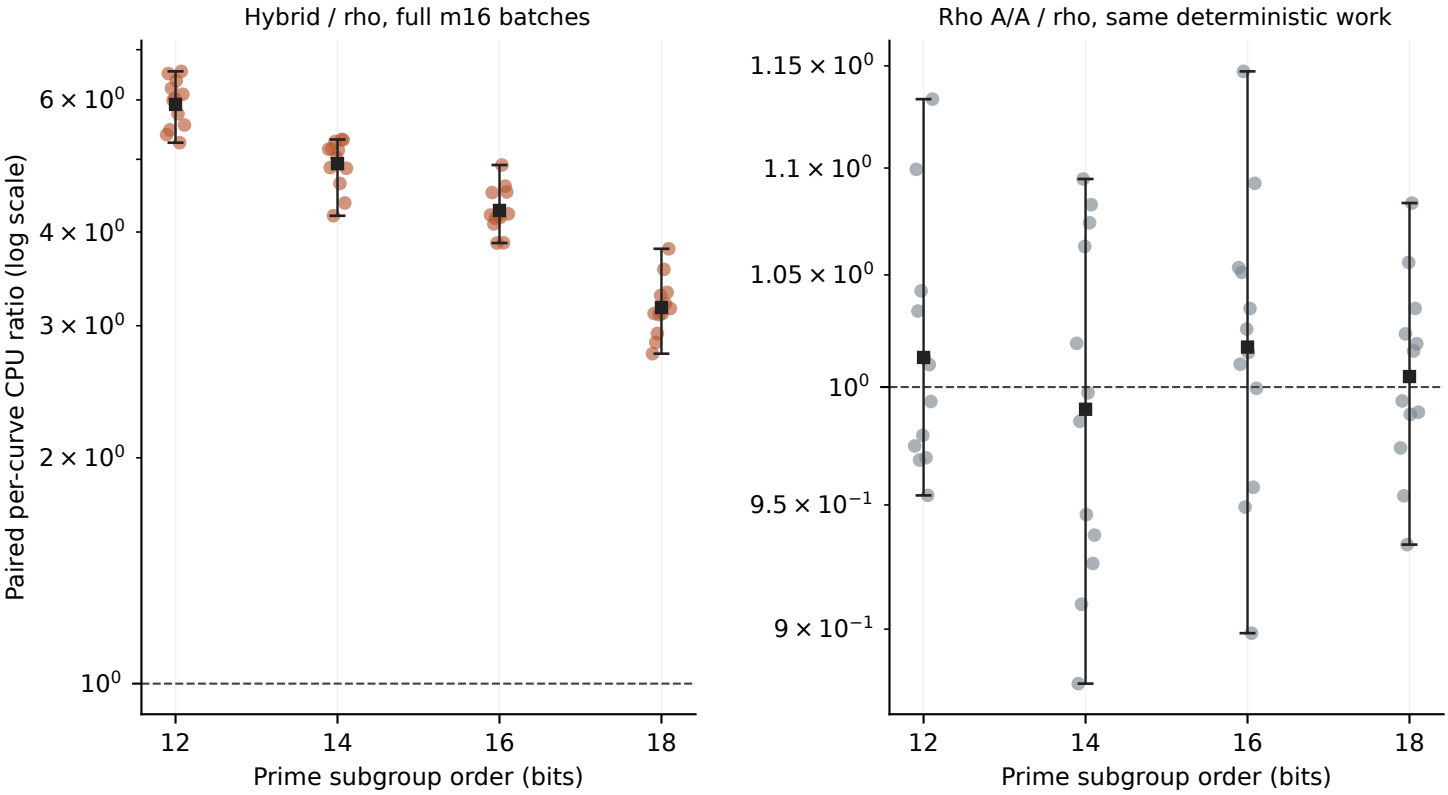
Each dot is one curve; lines are equal-curve geometric means. Left: setup plus mean target CPU, reconstructed as a cold-use cost. Right: setup amortized over 16 targets. These are not independently rerun cold targets.

Arm	Setup CPU(s)	Online CPU(s)	Aggregate ms/target
AutoRho	0	0.308699	0.401952
Rho A/A	0	0.310823	0.404718
Candidate + rho	0.275634	0.992687	1.65146
Bernstein-Lange	0.0536332	0.131367	0.240886
BSGS	0.00562228	0.0533561	0.0767948

The aggregate table divides summed measured setup plus target CPU by 768; it is not a mean of paired ratios. Shared curve/target validation, parsing, JSON, orchestration and final teardown are excluded from arm timers and reported separately. Full per-curve data retain all statuses.

Matched paired costs and A/A

Every point is a curve; complete positive-time pairs only



Bits	Hybrid/rho	BL/rho	BSGS/rho	A/A/rho	Hybrid/BL	BSGS/BL
12	5.92 (12/12)	0.649 (12/12)	0.114 (12/12)	1.01 (12/12)	9.12 (12/12)	0.176 (12/12)
14	4.93 (12/12)	0.407 (12/12)	0.15 (12/12)	0.99 (12/12)	12.1 (12/12)	0.368 (12/12)
16	4.27 (12/12)	0.521 (12/12)	0.195 (12/12)	1.02 (12/12)	8.2 (12/12)	0.373 (12/12)
18	3.17 (12/12)	0.62 (12/12)	0.224 (12/12)	1 (12/12)	5.11 (12/12)	0.362 (12/12)

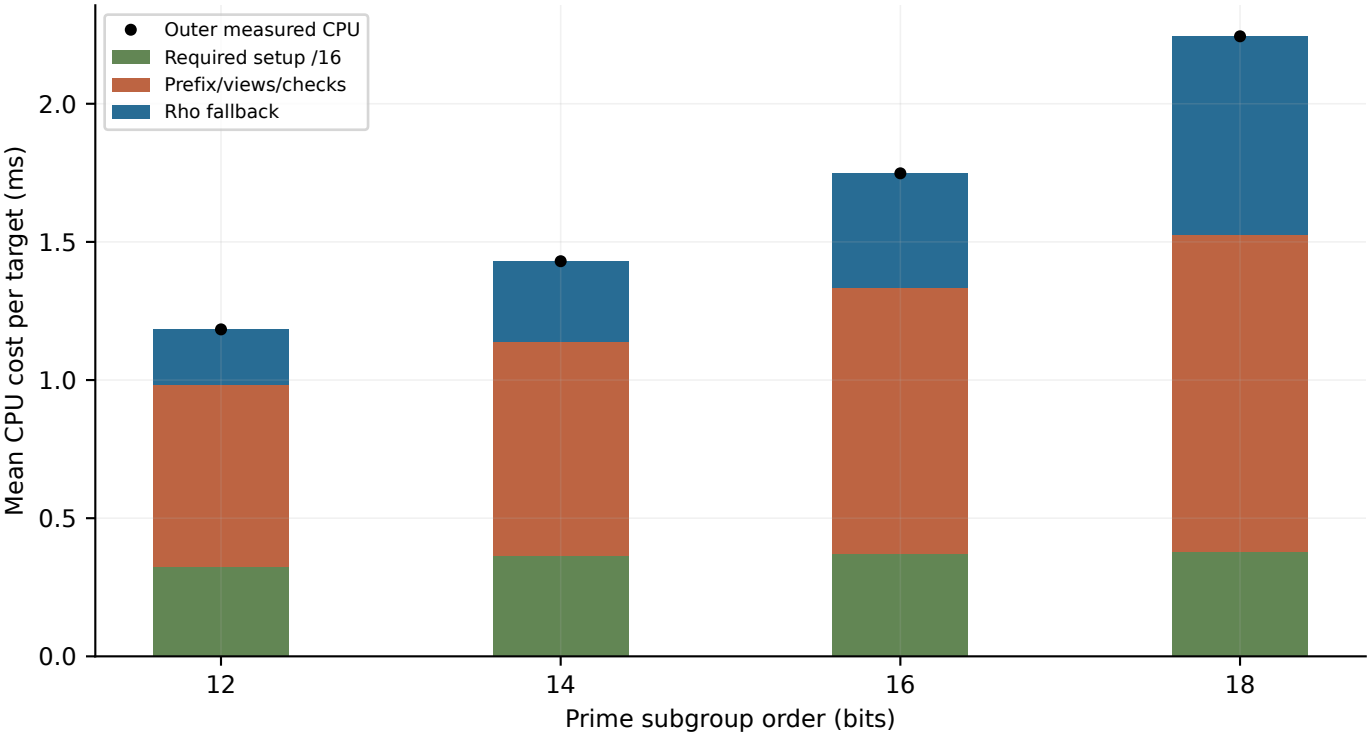
Table entries are per-size geometric means with comparable/requested curve counts. Dots retain all valid curve ratios; squares and whiskers show the geometric mean and complete range, not confidence intervals. A/A uses identical seed, returned scalar and deterministic work.

A/A work/answer equality holds for 768 pairs. There are 0 target pairs with unresolved zero timing. Censored or invalid pairs remain in the outcome counts; their successful-only ratios cannot establish a full-panel gain.

The BL-referenced columns are additional descriptive arithmetic; the registered primary comparator remains rho. Ratios describe this revision, toy sizes and reuse regime. No bootstrap, p-value or asymptotic exponent fit was computed.

Where the primary hybrid spends work

Prefix work, charged setup and exact-seeded rho fallback



Bars show recorded stage CPU per target with setup amortized over 16; dots show the outer hybrid CPU envelope. Small differences reflect wrapper/timing boundaries. Views, inference and candidate checks are nested inside prefix CPU, not extra additive stages.

Stage	Group adds	Field multiplies	Scalar calls	DP terms
Setup	393,201	21,801,946	98,294	0
Prefix	1,087,180	44,585,586	9,216	10,260,480
Fallback	570,098	22,694,956	18,242	0

Scalar multiplication invokes group operations, and powers/inversions invoke field multiplies. These call counts are not disjoint work units and must not be added. Full process CPU also captures the candidate DP/log/sort and upstream table/hash work omitted by the field-operation vector.

Memory values are partial kernel/vector payloads and upstream byte estimates. Measured per-arm peak RSS is unavailable; the 12GiB operational limit is not measured memory consumption.

Complete secondary confirmation matrix

Descriptive bounded-candidate outcomes; no post-hoc winner selection

Feature	Mode	Beam	Hits/cases	Checks	Sum B/r
constant	model	1	0/768	768	0.0962
constant	model	8	0/768	6144	0.7694
constant	exact	1	0/768	768	0.0962
constant	exact	8	0/768	6144	0.7694
x hash 7	model	1	0/768	768	0.0962
x hash 7	model	8	0/768	6144	0.7694
x hash 7	exact	1	0/768	768	0.0962
x hash 7	exact	8	2/768	6144	0.7694
x mod 7	model	1	0/768	768	0.0962
x mod 7	model	8	2/768	6144	0.7694
x mod 7	exact	1	1/768	768	0.0962
x mod 7	exact	8	3/768	6144	0.7694
fifth residue	model	1	0/480	480	0.0621
fifth residue	model	8	0/480	3840	0.4967
fifth residue	exact	1	0/480	480	0.0621
fifth residue	exact	8	3/480	3840	0.4967

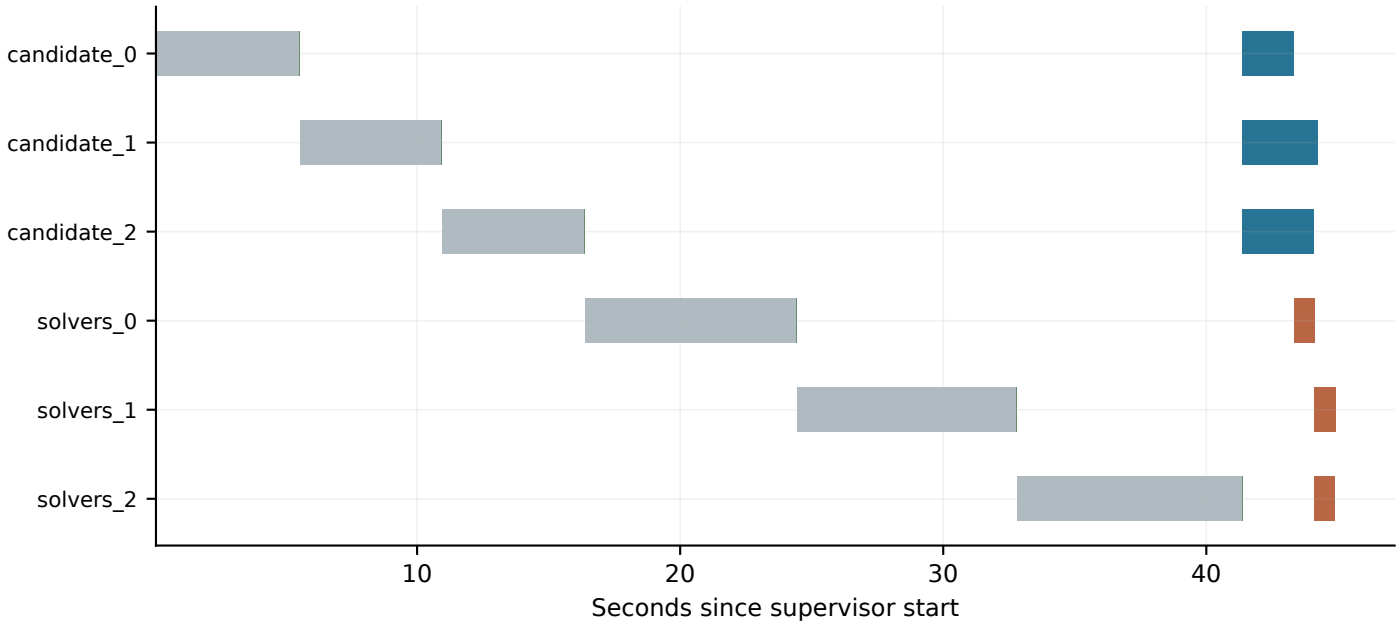
The secondary study has 174 kernels, 11,136 target cases and 18 structural NAs. The primary x-mod-7 exact beam 8 lists match their secondary counterparts. Counting both would duplicate the same candidate answers.

B/r is an ideal-uniform guessing expectation for the constant control and only a candidate-budget reference for nonconstant features. Constant modes can duplicate guesses; reused targets/configurations are not independent curve observations.

Training, heldout calibration, shared views and all other configurations remain charged study work. They are not silently added to or subtracted from the separately executed primary algorithm.

Execution, custody and practical limits

Raw observations are published before this report



Gray: build; green: fixed self-test; blue/orange: measurement jobs. Up to three jobs run concurrently. These intervals describe execution and do not replace the paired per-arm CPU measurements.

The receipt reports output_validated. Hash checks cover 2882 receipt artifacts, 2560 producer slots, all six original byte streams and 15744 aligned independent verification cases. All recorded candidates/final answers were compared with separately coded Jacobian results.

Public scalar labels and generation seeds were excluded from solver/checker inputs. This is source-enforced separation, not OS-isolated blindness. The source revision, source/input digests, exact commands and byte-level parts index are linked in HEADER.md and data/summary.json.

The 48 curves cover four small subgroup-bit levels. The nested core overlaps the full panel; full versus core reuse must remain explicit. Observed CPU scaling and differences at these inputs do not establish an asymptotic exponent. Working correctness and incremental comparison are distinct from novelty and later independent review.

Retained data identify the actual dominant stages, fallback dependence and table/reuse costs for the next implementation iteration. No mathematical direction is closed by this finite cohort.